# JICHANG YANG

*Department of Electrical and Computer Engineering*
*The University of Hong Kong (HKU)*
✉ *yangjc100@connect.hku.hk*
*yangjc97.github.io*


## Academic Appointment

Sept. 2026 – Present **Postdoctoral Fellow**, *Department of Electrical and Computer Engineering*, HKU.
Advisor: **Prof. Han Wang**
Research: In-memory computing, edge AI, and hardware-software co-design.

## Education

Sept. 2022 – Aug. 2026 **Ph.D.** in Department of Electrical and Computer Engineering (ECE)
The University of Hong Kong (HKU), Hong Kong, China (GPA: 4.1/4.3)
Advisors: **Prof. Han Wang** and **Prof. Zhongrui Wang**
Research Interests: In-memory computing, analog circuits, and resistive memory

Sept. 2019 – Jun. 2022 **M.S.** in School of Electrical and Electronic Engineering (SEEE)
Huazhong University of Science and Technology, Wuhan, China (GPA: 3.97/4.0)
Advisors: **Prof. Dong Jiang** and **Prof. Ronghai Qu**
Research Interests: Active magnetic bearings, control systems, and power electronics

Sept. 2015 – Jun. 2019 **B.S.** in School of Electrical and Electronic Engineering (SEEE)
Huazhong University of Science and Technology, Wuhan, China (GPA: 3.95/4.0)
Major: Electrical Engineering and Automation (Elite Class)

## First Author & Co-First Publications

† indicates equal contribution; * indicates corresponding author

### Journal Articles

[J.1] 2026 **Jichang Yang**†, Hegan Chen†, Jia Chen†*, Songqi Wang, Qifan Zhu, Xinyuan Zhang, Yan Zeng, Shaocong Wang, Mingrui Yang, Yifei Yu, Xi Chen, Bo Wang, Binbin Cui, Yi Li, Ning Lin, Meng Xu, Yi Li, Xiaoxin Xu, Xiaojuan Qi, Xumeng Zhang*, Dashan Shang*, Zhongrui Wang*, Han Wang, Qi Liu, Kwang-Ting Cheng, and Ming Liu. Resistive memory-based neural differential equation solver for score-based diffusion model. ***Nature Communications*** *(Impact Factor: 18.1)*, 2026.

[J.2] 2026 Yi Li†, **Jichang Yang**†, Qunsheng Hou†, Songqi Wang, Sishuo Liu, Zijian Ye, Jizhong Jiang, Ning Lin, Xiaoxin Xu, Dashan Shang, Xiaojuan Qi, Zhongrui Wang*, and Han Wang. Adaptive redox resistive memory programming for efficient and robust class-incremental learning. ***Advanced Materials*** *(Impact Factor: 29.1)*, 2026.

[J.3] 2025 Hegan Chen†, **Jichang Yang**†, Jia Chen†*, Songqi Wang, Shaocong Wang, Dingchen Wang, Xinyu Tian, Yifei Yu, Xi Chen, Yinan Lin, Qifan Zhu, Yangu He, Xiaoshan Wu, Yi Li, Xinyuan Zhang, Ning Lin, Meng Xu, Yi Li, Xumeng Zhang, Xiaojuan Qi, Zhongrui Wang*, Han Wang*, Dashan Shang*, Qi Liu, Kwang-Ting Cheng, and Ming Liu. Continuous-time digital twin with analog memristive neural ordinary differential equation solver. ***Science Advances*** *(Impact Factor: 13.9)*, volume 11, page eadr7571, 2025.

[J.4] 2021 **Jichang Yang**, Dong Jiang*, Hongbo Sun, Jianfu Ding, An Li, and Zicheng Liu. A series-winding topology converter with capability of fault-tolerant operation for active magnetic bearing drive. ***IEEE Transactions on Industrial Electronics*** *(Impact Factor: 7.4)*, volume 69, pages 6678–6687. IEEE, 2021.

[J.5] 2020 **Jichang Yang**, Dong Jiang*, Hongbo Sun, An Li, and Zicheng Liu. Series-winding topology converter for active magnetic bearing drive. ***IEEE Transactions on Industrial Electronics*** *(Impact Factor: 7.4)*, volume 68, pages 11772–11782. IEEE, 2020.

### Conference Papers

[C.1] 2026 **Jichang Yang**†, Xinyuan Zhang†, Songqi Wang†, Hegan Chen, Zimo Li, Shuang Liang, Hang-Ming Zhang, Ning Lin, Yi Li, Zhongrui Wang*, and Han Wang*. All-PWM RRAM compute-in-memory with in-loop state memory for energy-efficient closed-loop state evolution. In ***2026 IEEE International Electron Devices Meeting (IEDM)***, 2026. Accepted.

[C.2] 2026 **Jichang Yang**†, Xinyuan Zhang†, Jia Chen†, Meng Xu, Hegan Chen, Yan Zeng, Jizhong Jiang, Songqi Wang, Yi Li, Zhongrui Wang*, Yi Li*, and Han Wang*. Comprehensive reliability enhancement for high-precision RRAM compute-in-memory via fault-aware hadamard transform and low-rank compensation. In ***2026 IEEE International Electron Devices Meeting (IEDM)***, 2026. Accepted.

[C.3] 2024 **Jichang Yang**†, Hegan Chen†, Jia Chen, Songqi Wang, Shaocong Wang, Yifei Yu, Bo Wang, Ning Lin, Xinyuan Zhang, Rui Chen, Zhongrui Wang*, Dashan Shang*, Han Wang*, Qiang Liu, and Ming Liu. Conditional diffusion model acceleration with first-demonstrated rram-based in-memory neural differential equation solver. In ***2024 IEEE International Electron Devices Meeting (IEDM)***, pages 1–4. IEEE, 2024.

[C.4] 2026 Kwunhang Wong†, **Jichang Yang**†, Karl M. H. Lai, Hegan Chen, Songqi Wang, Wei Xuan, Ning Lin*, Han Wang, Xiaojuan Qi*, and Zhongrui Wang*. RRAM-DP: Device-calibrated differential privacy for in-memory edge learning. In ***2026 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)***, 2026. Accepted.

[C.5] 2026 Yangu He†, Xinyuan Zhang†, **Jichang Yang**†, Yue Zhang, Kwunhang Wong, Bo Wang, Ning Lin*, Xiaojuan Qi*, Zhongrui Wang*, and Han Wang*. NeuronUnlearn: Machine unlearning for RRAM-based computing-in-memory accelerators. In ***2026 44th IEEE International Conference on Computer Design (ICCD)***, 2026. Accepted.

[C.6] 2026 Songqi Wang†, Meng Xu†*, **Jichang Yang**†, Zhexu Chen, Hegan Chen, Sishuo Liu, Xinyuan Zhang, Kwunhang Wong, Ning Lin, Zhongrui Wang*, and Han Wang*. A fully analog continuous-time cim neural ode solver for flow-matching-based fluid dynamics generation. In ***2026 63rd ACM/IEEE Design Automation Conference (DAC)***, pages 1–7. IEEE, 2026.

[C.7] 2025 Ning Lin†, Yi Li†, Jiankun Li†, **Jichang Yang**†, Yangu He, Yukui Luo, Dashan Shang, Xiaoming Chen*, Xiaojuan Qi*, and Zhongrui Wang*. Guarder: A stable and lightweight reconfigurable rram-based pim accelerator for dnn ip protection. In ***2025 62nd ACM/IEEE Design Automation Conference (DAC)***, pages 1–7. IEEE, 2025.

## Research Experience

### Department of Electrical and Computer Engineering, HKU

Apr. 2024 – Present — ***High-Precision Analog Computing Using Resistive Memory***.

- Developed high-precision RRAM circuits for analog computing, mitigating device non-idealities including write noise and stuck faults.
- Designed an all-PWM RRAM compute-in-memory architecture with in-loop state memory, enabling energy-efficient closed-loop state evolution without in-loop ADC/DAC conversions.

### In-Memory/Sensor Computing Lab, HKU and Intelligent Agents Lab, SUSTech

Sept. 2022 – Present — ***Resistive Memory-Based Neural Differential Equation Solver***.

- Designed custom PCBs and performed experimental debugging of RRAM macros to validate analog computing performance in hardware.
- Implemented a fully analog neural differential equation solver for score-based diffusion models, demonstrating improved generation speed and energy efficiency.
- Co-designed an RRAM-based hybrid neural differential equation solver and implemented algorithm deployment for large-scale AI tasks.

Center for Advanced Electrical Machines and Drives, HUST

Oct. 2018 – Jun. 2022 ***Series-Winding Topology Converter for Active Magnetic Bearing Drive***.

- Built a universal active magnetic bearing controller for high-performance real-time control.
- Proposed a series-winding topology for active magnetic bearing converters, improving DC voltage utilization while reducing the active device count.
- Devised fault-tolerant control strategies to sustain converter operation under switching-device open-circuit failures.

## Teaching Assistant

Fall 2025 ELEC3350 Electronic Circuits and Devices I, Faculty of Engineering, HKU

Fall 2024 ELEC3350 Electronic Circuits and Devices I, Faculty of Engineering, HKU

Spring 2024 ELEC6049 Digital System Design Techniques **(Best TA Award)**, Faculty of Engineering, HKU

Spring 2023 ELEC6049 Digital System Design Techniques, Faculty of Engineering, HKU

## Fellowships & Awards

2022–2026 **Hong Kong PhD Fellowship (HKPF)** and HKU Presidential PhD Scholarship (HKU-PS)

2024 Best Poster Award in 2024 Nature Conference on Neuromorphic Computing

2024 HKU-EEE Teaching Assistant Award

2022 Y. S. and Christabel Lung Postgraduate Scholarship

2022 Outstanding Graduate, HUST

2021 **National Scholarship for Postgraduates**, 2020–2021 Academic Year

2020 **National Scholarship for Postgraduates**, 2019–2020 Academic Year

2019 Outstanding Undergraduate Graduate, HUST

2018 Outstanding Undergraduate Student (Special Distinction), HUST

## Technical Skills

Intelligent Software: AI model development, algorithm implementation, and hardware-software co-design using Python.

Hardware Design: Mixed-signal circuit design, PCB prototyping, and analog hardware validation using Altium Designer and LTspice.

Embedded Platforms: Real-time system implementation and algorithm deployment on ARM, DSP, and FPGA platforms.

Modeling and Simulation: System-level modeling, numerical simulation, and performance verification using MATLAB and ANSYS.